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A Techno-Economic Comparison of Traditional Upgrades, Volt-Var Controls, and Coordinated Distributed Energy Resource Management Systems for Integration of Distributed Photovoltaic Resources

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Abstract – In some cases, increased deployment of distributed photovoltaic (PV) systems can impact the distribution grid, including causing steady-state voltages and thermal loading of lines and equipment to deviate from operational limits. We present techno-economic analysis of three possible solutions for mitigating these effects on two real feeders: traditional infrastructure upgrades, autonomous volt-var controls, and a distributed energy management system (DERMS). We focus on issues arising in aggregate from residential and small commercial PV systems at different penetration levels and use a bottom-up approach that couples cost modeling with sequential hosting capacity analysis and quasi-static time-series simulation. We compare trade-offs for each solution in terms of effectiveness, upfront capital costs, operating costs, PV output curtailment, and distribution system losses. We find that volt-var controls offer the lowest cost option for hosting capacity expansion but cannot mitigate all violations at high penetration levels. For the feeders studied, the range of upfront cost for the DERMS and traditional upgrades is similar, but the DERMS results in a greater expansion of the hosting capacity. However, unlike traditional solutions, DERMS also involves higher level of PV curtailment than is observed with autonomous volt-var controls.

Keywords – distributed solar, distributed energy management systems (DERMS), volt-var control, time-series analysis, cost analysis, hosting capacity

I. INTRODUCTION

The installation of distributed photovoltaic systems (DPV) can, in some cases, negatively affect the operating conditions on the distribution system, including causing over- or under-voltages, increased voltage fluctuation, thermal overloading on lines and transformers, or issues with protection coordination. DPV deployment has increased consistently since 2010 (SEIA and Wood Mackenzie Power & Renewables 2019), with particularly rapid increases on certain distribution systems or feeders. Even some regions of states like Missouri and South Carolina that are not top markets for distributed solar have seen exponential increases in the number of net energy metered (NEM) customers over the last 5 years (K. Horowitz et al. 2019). This has created a need to better understand how these resources impact the distribution system and holistically analyze different options for their integration in order to inform standards, policies, and distribution system planning.

The penetration level at which DPV systems cause operating violations on the distribution system is referred to as the *hosting capacity*. The hosting capacity is highly variable and depends on a number of factors, including:

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- The location of the DPV generators and other DERs on the feeder
- The time-varying load and generation profiles
- The feeder configuration and existing equipment on the feeder
- The controls on the DPV inverter and existing grid equipment

Additionally, hosting capacity is not a hard limit on the amount of DPV generation that can be incorporated, but rather the point at which an upgrade, including either new equipment or a control setting or operational change, is required to mitigate the identified issues. These upgrades incur a cost and then allow for additional DPV systems to be deployed without causing violations, i.e., the hosting capacity is now different (and higher) with appropriate upgrades.

Traditionally, distribution system violations at the hosting capacity have been mitigated using utility infrastructure upgrades, including, for example, replacing overloaded lines and transformers, installing new voltage regulating equipment, or adjusting the control settings on existing equipment, such as load tap changes (LTCs), line regulators, or capacitor banks. Utilities typically assess the impacts during the interconnection process and identify required upgrades. A number of prior studies have examined the potential costs and hosting capacity expansion potential of traditional upgrade options (Xu et al. 2019)(K. A. W. Horowitz, Palmintier, et al. 2018)(Bird et al. 2018)(Aziz and Ketjoy 2018). Studies consistently find that the costs depend significantly on the circuit characteristics as well as the location of DPV systems on the feeder, with the effects tending to be worse when PV installations are clustered far from the substation. Dynamically reconfiguring the circuit or encouraging DPV deployment on specific areas of the circuit can reduce operational impacts (Takenobu et al. 2019), although the latter is often challenging in practice giving legal and siting constraints.

Advanced inverter controls are another potential option for expanding the hosting capacity that allow for more flexibility in the DPV output. Such controls are now part of interconnection standards, including the current version of the IEEE 1547 Standard for the Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces (IEEE Standards Coordinating Committee 21 2018), California's Rule 21, and Hawaii Rule 14H. Advanced inverter functions have been utilized for an even longer amount of time in countries like Germany and parts of Australia that reached high DPV penetration levels earlier than the United States. The use of advanced inverters to control the real and reactive power output of DPV systems has also been identified as a potential alternative to traditional infrastructure upgrades for the integration of DPV systems that have been shown to provide enhancements in hosting capacity and reduced integration cost in some—but not all—scenarios and in different countries with different types of distribution systems (DNV GL, prepared for the California Public Utilities Commission 2018);(K. A. W. Horowitz, Ding, et al. 2018). In particular, thermal overloading cannot always be totally resolved depending on the advanced inverter settings and the fractions of DPV with advanced controls, and certain voltage issues may not be easily mitigated with advanced inverter controls if grid voltage regulators or capacitors have fixed (e.g. time-based) settings. In fact, (Procopiou and Ochoa 2019) found that in some cases the reactive power absorbed by PV with volt-var controls can increase thermal overloading on distribution assets. (Idlbi et al. 2013) considered case studies in Spain, Germany, and Slovenia with DPV and/or distributed wind and saw that in certain scenarios the total cost indication with traditional grid reinforcement was much higher than the use of advanced inverters and control strategies, while in other cases it was lower; the authors also found this relationship was very

sensitivity to the feed-in tariff values, which affect the value of the real power exported from distributed generators. Additional studies have found that the use of advanced inverters coordinated with traditional upgrades can further boost the hosting capacity (Ku et al. 2019)(Keane and O'Malley 2007).

One key aspect of understanding the commercial feasibility of utilizing advanced inverter functions to expand the hosting capacity is understanding the impact of these functions on the real power output of the DPV system (curtailment) and the economic implications of curtailment for the system owner. Estimates of curtailment and, in some cases, translation of these estimates to economic impact has been explored in international literature when volt-var control or coordinated control schemes are utilized[1](Lusis et al. 2019; Davidson and McArthur 2007; Dolan et al. 2009; Gill et al. 2015; Keane and O'Malley 2007; Ochoa et al. 2009). The greatest experience with these schemes internationally has been with larger distributed wind generators in Europe, where these solutions have been explored since the early 2010s.

Several recent papers have explored this question from a technical angle for distribution grids in the United States. A Pacific Gas & Electric (PG&E) study under the Electric Program Electric Charge (EPIC) program modeled curtailment when two different volt-var curves were implemented on DPV systems in PG&E territory. The authors found variation in curtailment between 0.09% and 1.01% of annual energy curtailment with an average curtailment of 0.48% for DPV inverters in the field (PG&E 2019). They also modeled curtailment in additional scenarios and found that only 45 out of the 8,414 PV installations modeled had a curtailment greater than 1%. In a 2017 study for the Hawaiian Electric Company, researchers estimated curtailment in a similar range for a high-penetration scenario, with 95% of customers experiencing 0.5% or less in annual energy curtailment with a volt-var/volt-watt hybrid control (Giraldez et al. 2017). Some customers experienced higher curtailment, in the range of 1% to 5%; however, in a 2018 study the authors updated results to show that these high numbers were partly influenced by the algorithm used for convergence of the simulation and the specifics of the PV scenarios, noting that they expect maximum customer curtailment to be lower than they initially estimated (Giraldez Miner et al. 2018).

Similarly, an Electric Power Research Institute (EPRI) report estimated the amount of PV curtailment that would enable a 25% increase in the hosting capacity of DPV systems using different volt-var controls resulted in real power curtailment of less than 0.08% of annual energy production in all scenarios (EPRI 2019). EPRI found that this curtailment was economical in a wide range of scenarios compared to traditional grid upgrades but noted that this relationship will likely evolve as penetration levels increase. For example, beyond a certain penetration level, curtailment may become uneconomic or the specific advanced inverter functions may not mitigate all violations. This is a similar conclusion to that of (Medley 2018), who found a potential for large expansions in hosting capacity with advanced inverters, while keeping curtailment costs below those associated with traditional upgrades. More aggressive advanced inverter settings than the IEEE 1547-2018 curves were used, resulting in higher curtailment. Medley et. al also found that thermal load shifting could reduce the amount of curtailment with smart inverters. Finally, Medley et. al found that the solutions that are most commercially and technically viable are often very specific to a given system. The best solution is influenced by a

[1] These coordinated control schemes often referred to as active network management (ANM) in that literature.

variety of factors, including the existing infrastructure and capabilities on the feeder, the resources and preferences of the DPV system owner, and regulatory policy or constraints within a given region.
But, despite this rich literature, there is still a lack of techno-economic analysis of *coordinated dynamic integration strategies, such as DERMS*. This is particularly true on U.S. grids, since many existing studies focus on European systems that have a different structure. While (Medley 2018) does cover this topic, they looked exclusively at coordinated control strategies based on pre-defined principles of access, rather than optimal control, on larger DPV systems (> 500 kW). Large DPV systems have both different effects on the grid and can likely tolerate different levels of curtailment than smaller, customer-owned systems. An earlier study comparing advanced inverters alone to the use of a distribution management system (DMS) found that the DMS offered the greatest technical and economic benefits (Palmintier, Giraldez, et al. 2016), but it too looked at large-scale (5MW) DPV systems. This paper helps fill this gap by considering coordinated control systems for regions dominated by smaller-scale DPV.

Additionally, prior analyses on dynamic integration strategies consider only one or a few DPV deployment scenarios, and thus the potential trade-offs of different strategies for hosting capacity expansion under uncertain DPV growth is still unknown. Looking at a range of penetration levels is also important to help overcome reluctance to the adopt of advanced inverter controls due to uncertainty about how DPV curtailment could evolve in the future. Additionally, some impacts, such as losses and device operations, could initially see an increase but decrease later at higher penetrations. This paper also seeks to help understand how these change as a function of penetration level can help illuminate the overall cost-benefit of increasing DPV deployment and inform policy design and cost allocation approaches.

This paper seeks to address these gaps by providing these innovations:

- A comparative techno-economic analysis of traditional infrastructure upgrades, autonomous volt-var, and DERMS approaches for expanding the hosting capacity
- Analysis focused on two real distribution feeders in the United States
- Scenarios where circuits are dominated by small residential and commercial DPV systems
- Consideration of the costs, curtailments, and grid impacts that consider a richer set of costs and perspectives, such as customer curtailment costs, upfront capital cost of the DERMS system and all required communications infrastructure, which has not been previously included in the literature to our knowledge.
- Understanding of how curtailment risk could evolve under many different PV deployment scenarios as penetration levels grow on the circuit.

Our techno-economic analysis approach is bottom-up, based on one-year quasi-static time-series simulations (QSTS) in OpenDSS. As previous studies have also indicated that energy losses in the distribution system (Ma et al. 2019) and the number of grid device operations (Ghosh et al. 2019) can be affected by different DPV deployments, we also consider how these metrics change as a function of penetration, providing more broad analysis than other reports. However, the particular circuits considered in this analysis are regulated by switched capacitors only, so in this case device operations includes only to the number of capacitor switch operations.

II. MITIGATION STRATEGIES ANALYZED

We consider three different mitigation strategies in this work:

1) ***Traditional upgrades.*** We examine traditional upgrade strategies in combination with the use of alternative constant power factor settings on the inverter (in this case, we use power factor [PF] = 0.95 absorbing (lagging) for all scenarios). These include installation, removal, or transfer (to a new location) of equipment, including voltage regulators, and capacitor banks; reconductoring; and changing the control settings or set points on existing feeder equipment. These are implemented when the snapshot hosting capacity is reached. We previously determined traditional upgrades used for analysis in this paper as well as their associated cost as a function of penetration level (K. A. W. Horowitz, Ding, et al. 2018). The methodology for determining those upgrades involves a distribution engineering considering a set of typical upgrade solutions and simulating their ability to expand the snapshot hosting capacity. This method does not necessarily reflect an optimization approach, but rather is intended to reflect current interconnection practices.

2) ***Autonomous volt-var controls on all inverters.*** This is the scenario where inverters have autonomous volt-var control. We assume the IEEE 1547-2018 Category A volt-var curve, shown below in Figure *1*. We assume all inverters at all penetration levels have the volt-var controls in this analysis; in future work, we will explore the sensitivity of results to this assumption. The scenario where all inverters have advanced controls could be applicable to a case where little to no DPV generation has been deployed on the feeder prior to the implementation of the IEEE 1547-2018 and other advanced inverter standards, such that all new inverters might have these settings.

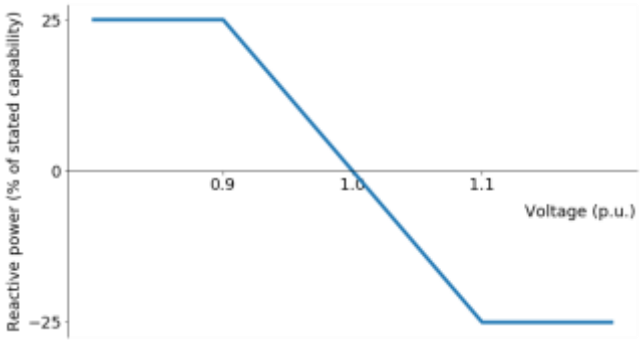


Figure 1. Default Category A volt-var curve from the IEEE 1547-2018 Standard applied in the mitigation strategy 2 scenario, wherein all inverters have autonomous volt-var controls.

Not all voltage violations could be mitigated at high penetrations using autonomous volt-var controls alone. We track the voltage impacts of DPV generation with volt-var on the system as a function of penetration and, when unacceptable voltage violations occur, assume traditional distribution grid upgrades would also be required.

3) ***DERMS.*** This case assumes a DERMS is used to send real and reactive power set points to the DPV systems on the feeder. This could represent a scenario where there are utility-owned and -operated rooftop DPV systems on a feeder, for example, as was done as part of Arizona Public Service's Solar Partner Program (Adhikari 2015), or a scenario where a

DER aggregator operates the DERMS. There are many possible scenarios for how this would develop in practice, and implementation will likely vary significantly over time, by location, and depending on the companies (utilities, DER aggregators) and other stakeholders involved. This paper attempts to provide an initial analysis of the techno-economic issues related to coordinated DERMS approaches for integration. The architecture implemented is shown in Figure 2. This assumes that all the DPV systems on the feeder at a given penetration level participate in the DERMS. In this algorithm, the P and Q are selected to optimize the output of the DPV generators (i.e., this is an optimization algorithm and not just a deterministic, coordinated control approach). Coordinated controls approaches that do not rely on solving optimization problems also have commercial traction and warrant further study. Our optimization algorithm heavily prioritizes settings that avoid direct reduction in real power (P). The DERMS operates only when there is a violation and has a maximum of 30 iterations per simulation to determine P-Q set points. The objective function, constraints, and other details of the DERMS algorithm used are included in the Appendix.

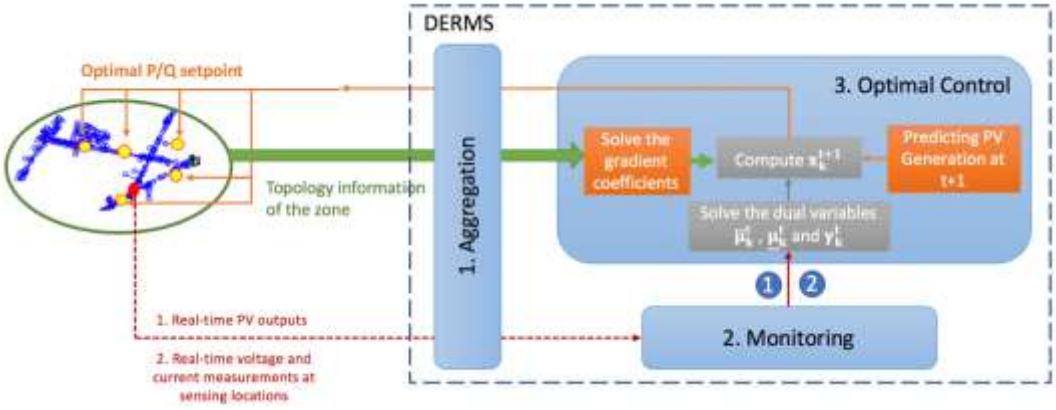


Figure 2. The architecture of the DERMS approach analyzed in this study. This approach has one controller per feeder that optimizes the P/Q setpoints of each DPV system on the feeder to avoid steady-state voltage and thermal violations.

For volt-var and DERMS, we assume that the PV inverter does not operate at night. It is now possible for some PV inverters to provide voltage regulation services during the night, and this is represents an interesting area for future research.

III. DEFINITION OF CURTAILMENT

We define *curtailment* as the reduction of real power output from the PV system due to a grid condition. For example, if a volt-var curve is applied to a PV inverter during a day where the real power output of the inverter is close to its rated value, real power may need to be curtailed to provide enough reactive power to regulate the voltage without exceeding internal inverter current limits. The amount of real power curtailment associated with providing a given amount of reactive power is related to the headroom for providing reactive power, or the inverter kVA-to-kW ratio. Curtailment can be avoided by oversizing the kVA of the inverter compared to the kW rating of the inverter.

This stands in contrast to *clipping*, which is the limitation of AC power output from the inverter below oversized DC panel output due to a DC-to-AC or inverter loading ratio greater than one. The DC-to-AC ratio, and thus amount of clipping, is determined by the designer of the PV system itself based on the system economics and the available inverters (e.g., an installer may only purchase inverters in a specific set of sizes, and the DC nameplate rating of the PV array may vary based on project or customer needs).

An example of curtailment under operation of an autonomous volt-var controller compared to clipping for an example PV system on a sunny summer day in Southern California is shown in Figure 3. Energy production lost due to clipping should not be counted as a grid integration cost or viewed as a barrier or concern for interconnection, since, as mentioned above, this is related to the PV system design itself and chosen by the PV system owner. The DC-to-AC ratio can influence both the degree of clipping and curtailment (EPRI 2019).

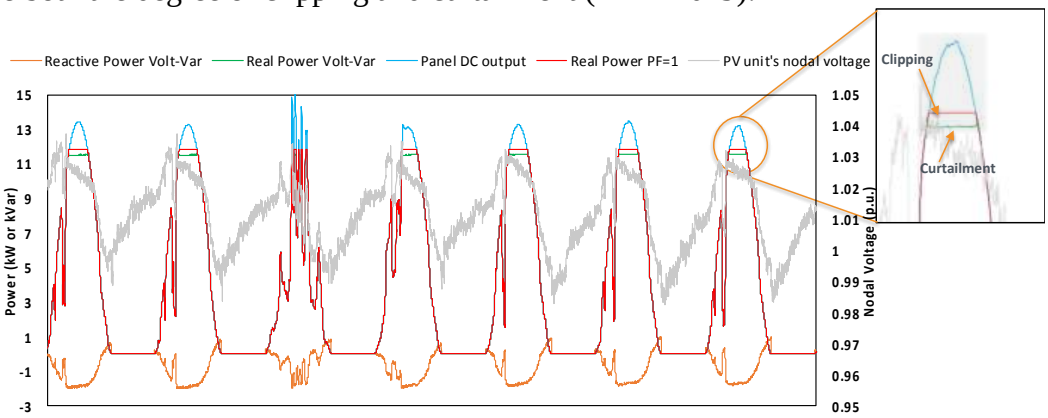


Figure 3. Illustration of the difference between curtailment resulting from the action of a volt-var controller versus clipping due to the DC-to-AC ratio for a PV system on a sunny summer day in Southern California.

Both clipping and curtailment stand in contrast to *tripping*, in which the PV system is disconnected or tripped off-line. Tripping could occur in cases with and without the use of advanced inverters; for example, a DPV system is tripped if voltages exceed 1.1 p.u., according to the IEEE 1547 Standard. Similarly, clipping may be present both with and without advanced inverter functionalities in use during high irradiance periods when the PV system is producing near its nameplate power. With unity power factor and no advanced inverter controls, however, no curtailment will occur. When advanced inverters are used to provide reactive power only, real power curtailment can still occur if the maximum possible reactive power provided with a given controller exceeds the headroom in the apparent (kVA) power rating of the inverter, causing real power to be reduced. This can be avoided by oversizing the kVA of the inverter to provide headroom. The potential economic trade-offs of doing so are included in the Results and Discussion section.

In addition to the characteristics of the PV panel and system, shading, weather, clipping, curtailment, and tripping all influence the expected or actual energy output of the DPV system in kWh. This energy output, coupled with the time at which it occurs and how the DPV system is compensated for power produced at different times, is ultimately the metric of most interest to the PV system owner. If, for example, using additional volt-var results in some curtailment but a

lesser amount of tripping, resulting in overall greater energy production from the PV system, that may be a preferable solution.

Curtailment is estimated for the scenarios with volt-var controls and DERMS controls. We first conduct a simulation with PF equal to one for a given DPV deployment and penetration and then simulate the same deployment and penetration with the advanced inverter controls. The total amount of annual curtailed energy is then calculated as:

$$E_{curtailed}(kWh) = \frac{\sum_t P_{PF=1}(t) - P_{advanced\ inverter}(t)}{60} \quad (1)$$

The sum is the sum of all time points (minutes) in the year. The divisor of 60 is needed to convert to kWh, since our simulations occur at a one-minute resolution.

When an advanced inverter controller operates, it may in practice step the DPV output (real or reactive power) up and down in smaller steps until it reaches the desired set point. In this analysis, we assume that the step up and down occurs in a symmetric fashion and cancels out, and thus the total amount of energy curtailed is not affected. We also assume that the latency of the response of the inverters for the coordinated (DERMS) and uncoordinated (volt-var) advanced inverter scenarios was less than that of our simulation, so it is not considered in calculating curtailed power. Indeed, response times are typically on the order of seconds for these functions (PG&E 2018).

IV. ANALYSIS METHODOLOGY

For this study, we look at two real feeders in Southern California with different characteristics, summarized in Table 1. The actual names and locations of the feeders are not specified to protect sensitive information. As can be observed from the table, these feeders have very different characteristics, so we would expect to see different effects of DPV generation on the distribution systems. Both Feeder A and Feeder B do not have real secondary network data in their electrical models. Instead, idealized star secondaries of 125 ft are used.

Using OpenDSS, we first conduct static hosting capacity analysis with a reduced set of stochastic deployments described below. We use the term *static hosting capacity* to refer to hosting capacity based on a few bounding, deterministic power flow conditions, including 1) maximum PV output and minimum daytime load, 2) maximum PV output and maximum daytime load, 3) no PV output and minimum daytime load, 4) no PV output and maximum daytime load. Static hosting capacity is the typical method of hosting capacity employed, contrasted with a time-series based or dynamic hosting capacity that looks at severity and duration of violations over a longer period of time (Mulenga, Bollen, and Etherden 2020)(Jain et al. 2019). We then conduct one-year QSTS power flow simulations on these feeders, which is used to calculate the annual energy losses in the distribution system; counts of reverse power flow; number of device operations (in this case, number of times the capacitor banks switch in a year since the circuits are capacitor-regulated); and reactive power at the substation. We also use the QSTS simulations at one-minute resolution to calculate the count of voltage and thermal violations when volt-var controls are used. For the control assumptions used in this paper, the DERMS was able to effectively mitigate voltage and thermal violations at all penetrations studied. One-minute

resolution was selected based on data availability. There is not sub-one-minute switching of capacitor banks on these feeders with their current control settings. Effects of DPV generation on protection coordination are not studied.

Table 1. Characteristics of the Feeders Used in This Analysis

Parameter	Feeder A	Feeder B
Primary voltage (kV)	12.47	4.16
Maximum load (MW)	7.65	1.99
Customers	625	250
Residential	291	220
Commercial	334	30
Maximum distance from substation to feeder node (kilometers)	9.7	2.36
Maximum X/R of primary bus	5.71	2.386
Minimum X/R of primary bus	0.669	0.7175
Average primary X/R	2.206	1.179
Capacitor count	6	2
Capacitor total kVAR	4800	600
Line regulator count	0	0
Substation load tap changer (LTC) count	0	0

DPV Deployments and Inverter Sizing Assumptions

In this work, we focus on residential and small commercial DPV system deployments, ranging in size from 1kW to 200kW, with the size of each PV system limited to < 2X the peak load. This algorithm was also used to generate DPV deployments in previous work (Ding et al. 2018), where the approach and resulting distribution of DPV system sizes for each case are described in more detail, but we briefly highlight key aspects of the DPV deployments scenarios here as well for reference.

We consider three DPV placements to capture a broad range of possible effects and costs of DPV integration, which can vary significantly with the location of DPV systems on the feeder as discussed above. These placements include:

- ***Far from the substation.*** DPV systems were clustered far from the substation. DPV systems located far from the substation tend to have more severe impacts on the distribution system.
- ***Close to the substation.*** DPV systems were clustered close to the substation. DPV systems clustered close to the substation tend to have less severe impacts on the distribution system.

- ***Random.*** DPV systems were randomly spread throughout the feeder. Randomly or evenly deployed DPV systems often, but not always, have impacts or costs that fall between scenarios where DPV systems are far from or close to the substation.

For each placement, we generate 10 samples, resulting in 30 total DPV deployments used in the analysis. The potential locations for DPV installation on each feeder for each placement are shown in Figure 4. As the diagram shows, for the far-from and close-to-the-substation placements, DPV systems are first deployed within the first 10% or last 10% of the feeder, respectively, until all the customers have a PV system, and then DPV systems may be deployed farther along toward the center of the feeder. Thus, while these three scenarios are initially distinct, they become more similar at high DPV penetrations because the DPV deployment spreads out along the length of the feeder in both the far-from and close-to-the-substation scenarios as customer penetration at either end of the feeder saturates. These three general scenarios have been used increasingly in the literature to capture a broad range of potential DPV impacts while reducing the number of simulations required compared to a full stochastic analysis (Ebe et al. 2017; Rylander, Smith, and Sunderman 2015; Palmintier, Broderick, et al. 2016). These spatial deployments were also used for all hosting capacity analysis included in our results.

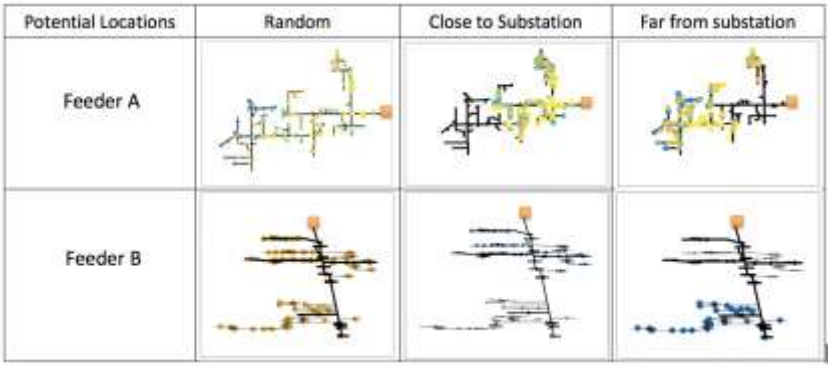


Figure 4. This analysis considers three potential scenarios of where DPV systems are sited on each feeder, with 10 different spatial deployments of DPV systems for each scenario (30 total DPV system locations considered). These scenarios capture a wide range of possible effects of DPV systems.

We assume all inverters have a DC-to-AC rating of 1.15 (oversized solar array) and have an apparent power rating (kVA) equal to the real power AC rating (kW_{AC}), meaning that inverters are not oversized to provide reactive power.

COST AND IMPACT ANALYSIS

We consider costs that utilities, DPV system owners, or ratepayers may incur under current regulations. However, we do not exhaustively consider all possible costs and benefits associated with DPV technology. Table 2 specifies which costs and benefits are included in our study. Distribution upgrades and associated costs are determined as described in (K. A. W. Horowitz, Ding, et al. 2018). The ranges of input unit cost data are from the NREL Distribution Grid Integration Unit Cost Database ("Distribution Grid Integration Unit Cost Database | Solar Research | NREL" 2018). The data is from a variety of public sources, as well as primary interviews and data collection conducted by NREL and from interconnection studies. Based on

our interviews with inverter manufacturers, utilities, and developers, once advanced inverter standards and testing procedures are set, and if the advanced inverter settings are enabled at installation, there is not an additional upfront cost of an advanced inverter compared to a traditional inverter. In fact, most inverters are now capable of providing advanced functionality that just needs to be switched on. If the inverters are legacy inverters that need to be retrofitted, the costs would be considerably higher, and if the inverters need to have advanced settings enabled after installation there is some small cost, depending on if this can be done remotely or a truck roll is required (K. A. W. Horowitz, Ding, et al. 2018). The DERMS up-front capital costs include the total cost of installing the DERMS on the feeder, including software, hardware, communications equipment, engineering, and interoperability. Note that DERMS costs are more uncertain and likely to evolve in the future due to the early nature of the technology and the fact that these systems have only been deployed on a limited scale in the United States as part of pilot or demonstration projects. Additionally, how DERMS costs will scale as the size of projects increases is not well known. For example, with increasing numbers of systems participating, additional communication infrastructure may be required. Ideally, the cost of operations and maintenance (O&M) and annual service or support fees for the DERMS software should be accounted for in an overall techno-economic evaluation. However, good and consistent data on O&M costs for traditional infrastructure upgrades is not widely available, and DERMS maintenance and support costs are not yet well known either due to the relatively early stage of the technology.

Table 2. Summary of Costs and Benefits That Are Included and Not Included in Our Analysis

Impact	How accounted for in this analysis
Distribution upgrade costs required to mitigate voltage or thermal violations, including the cost of new equipment or changing controls on existing equipment	• ***Traditional upgrade scenario*:** Up-front capital costs for upgrades identified using snapshot hosting capacity and upgrade analysis. • ***Autonomous volt-var control scenario:*** Assume all inverters have volt-var controls added at the time of installation, with no additional capital cost associated with enabling those controls. At the penetration level where volt-var can no longer mitigate these violations, traditional upgrades required are assumed to be needed. • ***DERMS scenario:*** Ranges for the total up-front capital costs of deploying DERMS from interviews with DERMS vendors ("Distribution Grid Integration Unit Cost Database | Solar Research | NREL" 2018) included.
Energy losses in the distribution system (e.g., line and transformer losses)	QSTS simulations are used to calculate the annual energy losses in the feeder (in kWh) with PF = 1, volt-var controls, and DERMS controls. These are then translated into a $/year cost by multiplying this kWh loss value by the average wholesale pricing (¢/kWh) for the SP-15 electricity hub in Southern California in 2019 (EIA 2019). These losses and costs at each DPV penetration level are compared to the base case with no DPV generation.
Curtailment of PV output	Curtailment for each DPV system is calculated as described above. This is reported on its own and also translated into some illustrative costs for different ownership and compensation structures to aid in understanding the potential magnitude of these costs.

Cost of any required upgrades to protection equipment	Protection cost impacts are not explicitly modeled in this study. However, we record if there is reverse flow at the substation and then discuss the cost of different upgrades that utilities have sometimes undertaken to ensure protection coordination in these scenarios. Understanding protection impacts of DPV integration and associated costs is a complex area of ongoing study.
Value of DPV to defer infrastructure investments or distribution system upgrades	Beyond the scope of this study and not included. However, this is an important factor to consider when looking at the overall cost-benefit ratio for DPV deployment in a particular utility region.
Any other value associated with DPV, including ancillary services	Beyond the scope of this study and not included. However, these could be important factors to consider when looking at the overall cost-benefit ratio for DPV deployment in a particular utility region.
Interconnection facilities costs (e.g., the equipment to physically and electrically connect a single DPV generator to the grid)	We categorize these as part of the DPV installed system cost, rather than a distribution grid integration cost, so they are not included in this study.
Interconnection study costs	Generally beyond the scope of this study; however, some engineering and study costs (e.g., those associated with determining new regulator set points) are included the unit cost data used in this study and thus represented.

For the cost of increased switching of capacitor banks, we look at the increase in switching operations compared to cases with no DPV generation and with PF = 1 (compared to advanced inverters). We then compare the number of switching operations to the estimated lifetime of the capacitor switch in number of operations.

For this analysis, we assume a 30,000-operation lifetime for the capacitor switches ("Distribution Grid Integration Unit Cost Database | Solar Research | NREL" 2018). However, capacitor switch lifetime can vary significantly due to environmental conditions (i.e., lifetime is determined by a variety of factors other than the number of switching operations or even loading on the device).

For the cost of energy losses in the distribution system (i.e., line losses, transformer losses, and losses in other distribution system components) or the cost of power demanded from transmission, we calculate a cost of losses based on the price of wholesale power that would be purchased. We take the amount of losses for a given scenario in kWh/year and multiply that by the ¢/kWh wholesale pricing to get a $/year cost. While this cost is not currently accounted for as part of the interconnection process or tracked as an explicit cost, it is a potential distribution system related cost (or benefit, if losses are reduced, as is the case in some scenarios), so we include it here using some example wholesale pricing for California. We also provide information on the kWh amount of losses as an alternative metric.

Because we assume that the inverters are not oversized to provide reactive power (kVA-to-kW ratio of the inverter is one for all cases), we do not include a cost for oversizing the inverter.

We look at cost as a function of penetration but do not explicitly predict the deployment of DPV systems over time (i.e., the penetration is not coupled to time). This makes it challenging to conduct full cash flow analysis, which must have information on when (in time) different DPV systems are deployed and different upgrades would be required. Because of this, we focus

instead on capital cost, annual cost of curtailment or losses, and other metrics (number of capacitor switch operations and thus potential early wear-out of capacitor switches). For utility-specific project planning, the expected DPV deployment over time, cost of capital for a given company, and other financial and regulatory factors should also be considered.

V. RESULTS AND DISCUSSION

Table 3 summarizes the up-front costs, ability to expand the hosting capacity, and the DPV curtailment for the traditional upgrades, autonomous volt-var, and DERMS cases. (K. A. W. Horowitz, Ding, et al. 2018) provide details of the upgrades required for the traditional upgrade scenario. The hosting capacity expansions shown are based on the static or snapshot hosting capacity methodology, which considers only several snapshot, worst-case power flow conditions and is the traditional approach used for planning, and interconnection. Higher PV penetration beyond the static hosting capacity may actually be able to be hosted due to the conservative nature of these limits; for example, traditional static approaches assume no Range A voltage violation can ever occur, while in reality the American National Standards Institute (ANSI) standard allows for a small amount of voltage outside of Range A for a limited duration, and these types of voltages are often observed in the field (Jain et al. 2019).

The capital cost per watt depends on the penetration level considered. For example, the cost of a DERMS solution is very high per watt if you only have one or two PV systems on the feeders, particularly if the systems are small. In that case, the cost would not likely be worthwhile.

Table 3. Summary of Up-front Costs, Ability to Expand Hosting Capacity, and DPV Curtailment for the Different DPV Integration Options Analyzed

	Feeder A			Feeder B		
	Traditional Upgrades Plus Inverter PF = 0.95 in Watt Priority	Autonomous Volt-Var Only	DERMS	Traditional Upgrades Plus Inverter PF = 0.95 in Watt Priority	Autonomous Volt-Var Only	DERMS
Up-front capital cost for upgrades to mitigate steady-state voltage and thermal violations ($)	$460,438–$493,772	$0	$300,000–$500,000	$157,333 - $491,022	$0	$300,000–$500,000
Increase in the static, snapshot hosting capacity	• **Random:** From 30.9% of peak load to 64.6% of peak load • **Far:** From 16.8% of peak load to 93.1% of peak load • **Close:** From 28.5% of peak load to 82.9% of peak load	• **Random**: From 30.9% of peak load to 60.8% of peak load • **Far:** From 16.8% of peak load to 61.5% of peak load • **Close:** From 28.5% of peak load to 53.6% of peak load	Until curtailment becomes uneconomical (voltage and thermal violations mitigated for all penetrations in our simulations)	• **Random**: From 23.1% of peak load to 27.8% of peak load • **Far**: Could not increase economically • **Close:** From 48.8% of peak load to 96.1% of peak load	• **Random**: From 23.1% of peak load to 24.6% of peak load • **Far**: No increase • **Close**: From 48.8% of peak load to 53.6% of peak load	Until curtailment becomes uneconomical (voltage and thermal violations mitigated for all penetrations in our simulations)
Total annual		Lower	Lower		Lower	Lower

curtailment for all PV systems on the feeder with worst deployment (%)	0%	penetration: • **Random at 37.9% of peak load:** 0.21% • **Far at 21.9% of peak load:** 0.21% • **Close at 31.1% of peak load:** 0.22% Higher penetration: • **Random at 131% of peak load:** 0.24% • **Far at 117% of peak load:** 0.23% • **Close at 130% of peak load:** 0.24%	penetration: 0% for all placements Higher penetration: • **Random at 131% of peak load:** 0.41% • **Far at 117% of peak load:** 4.02% • **Close at 130% of peak load:** 0.47%	0%	penetration: • **Random at 34.0% of peak load:** 0.17% • **Far at 6.3% of peak load:** 0.19% • **Close at 45.9% of peak load:** 0.23% Higher penetration: • **Random at 133.9% of peak load:** 0.39% • **Far at 94.6% of peak load:** 0.82% • **Close at 119.8% of peak load:** 0.33%	penetration: • **Random at 34.0% of peak load:** 0.97% • **Far at 6.3% of peak load:** 1.45% • **Close at 45.9% of peak load:** 0.15% Higher penetration: • **Random at 133.9% of peak load**: 7.14% • **Far at 94.6% of peak load:** 18.17% • **Close at 119.8% of peak load:** 3.19%

However, as the PV penetration level grows, the cost per watt becomes lower than the cost per watt for traditional upgrades. At the static hosting capacity (30.9% of peak load) and 133% of peak load on Feeder A (maximum penetration we simulated), for example, the median DERMS solution costs would be $0.16/W and $0.04/W, respectively. For comparison, for the random PV placement, the traditional upgrades on that feeder cost $0.19/W at the hosting capacity and $0.10/W at the new hosting capacity (64.6% of peak load). A DERMS solution could also potentially be used to achieve other system objectives and improve overall visibility and can allow for much higher PV penetrations to be accommodated without additional costs. In fact, the hosting capacity is expanded until curtailment becomes uneconomical for the PV systems participating in the DERMS and other upgrades are needed to bring curtailment to an acceptable level.

We observe no increase in hosting capacity for the far placement on Feeder B with volt-var because the feeder was limited by thermal overloading violations for that set of PV locations, which the volt-var was not effective at mitigating. For the traditional upgrade case, very expensive reconductoring and transformer replacement could be undertaken to mitigate those violations, but the expansion in hosting capacity is small, so these mitigations are uneconomical and viewed as unrealistic. In that case, it is likely that the DPV interconnection would not move forward in those locations.

The curtailment for all cases shown is calculated at the penetration level at which the first violation occurred on the feeder for a given placement (worst of all 10 deployments) as well as at the maximum penetration level simulated. Figure 5 visually shows the spread in curtailment among different generators for each advanced inverter control scenario and each feeder. The curtailment calculated is much lower with the volt-var control compared to the DERMS, but only a limited expansion in the hosting capacity is achieved, meaning that additional traditional

upgrades will also be required to further expand the hosting capacity to the highest penetrations modeled in this study (> 100% of peak load). For example, Figure 6 shows how the number of Range A voltage violations become large at high penetration levels even with the volt-var control applied.

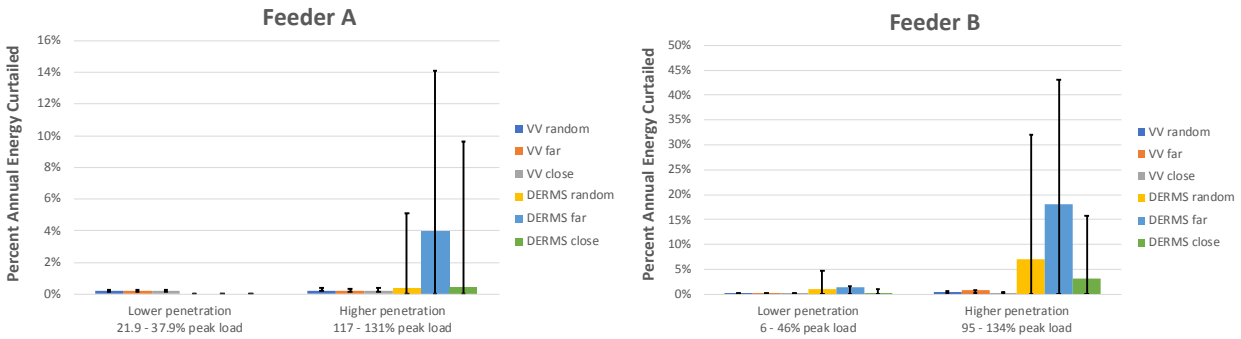


Figure 5. Comparison of the spread in curtailment for each generator on Feeder A (left) and Feeder B (right) with different controls approaches, PV placements, and penetration levels. VV = volt-var.

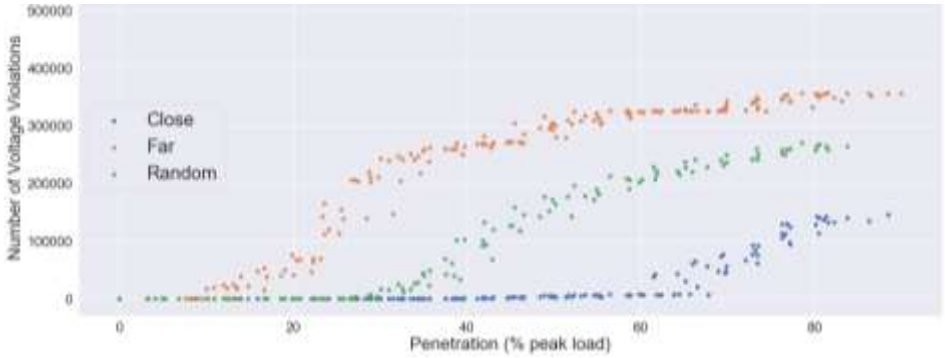


Figure 6. Increase in the modeled count of Range A voltage violations for all buses on the feeder during a one-year period, including under or overvoltage violations, when volt-var controls are used for Feeder B. This shows that the number of Range A voltage violations become large at high penetration levels even when volt-var controls are used.

The DERMS, in contrast, effectively mitigates the violations up until the maximum penetration simulated, but results in a higher level of curtailment overall and a larger spread in the curtailment experienced by different DPV systems. The curtailment is particularly high at higher penetration levels with the DERMS for Feeder B because of the thermal violations that arise, causing the DERMS to reduce the real power output from the DPV systems. Because thermal violations only occur on some of the transformers on the system, some PV systems experience very high curtailment whereas others experience none. As mentioned above, the use of volt-var controls alone cannot resolve these thermal overloading issues. Finally, as discussed in the Methods section, a DERMS has a risk of additional curtailment during loss of communications with the central controller/server, depending on the fail-safe approach in the controls. A 5% downtime in communications could contribute up to an additional 5% of real PV power production lost, depending on whether downtime overlaps with periods where DPV output would be curtailed some anyway. It is also possible that utilities will require an additional operating margin for the DERMS beyond the ANSI standard limits used in this study, and/or

may trip the generators off-line if they exceed a certain threshold. These requirements, if present, could result in additional curtailment in the DERMS scenario and thus warrant additional study in the future.

Figure 7 and Figure 8 are violin plots[2] that show how the distribution of curtailment among the all of the DPV systems evolves as penetration levels increase in the case of volt-var controls. We only simulate two penetration levels for the DERMS scenario due to the more computationally intensive nature of those simulations. The spread in curtailment experienced by each DPV system is critical to understand, as curtailment may be much larger for some systems than others and evaluating only the total curtailment for all systems on the feeder masks these differences. If the curtailment for one generator is sufficiently high, and depending on (alternative) cost of traditional infrastructure upgrades that would be triggered by that system's interconnection, that DPV system owner may choose not to participate in the DERMS (if optional) or the compensation and charges associated with the DERMS project may need to be structured to accommodate these systems. These figures illustrate that the evolution of the distribution in curtailment levels is feeder- and placement-specific. While the difference in curtailment between generators can be large, curtailment is below 1% for all PV units and scenarios with the volt-var controls.

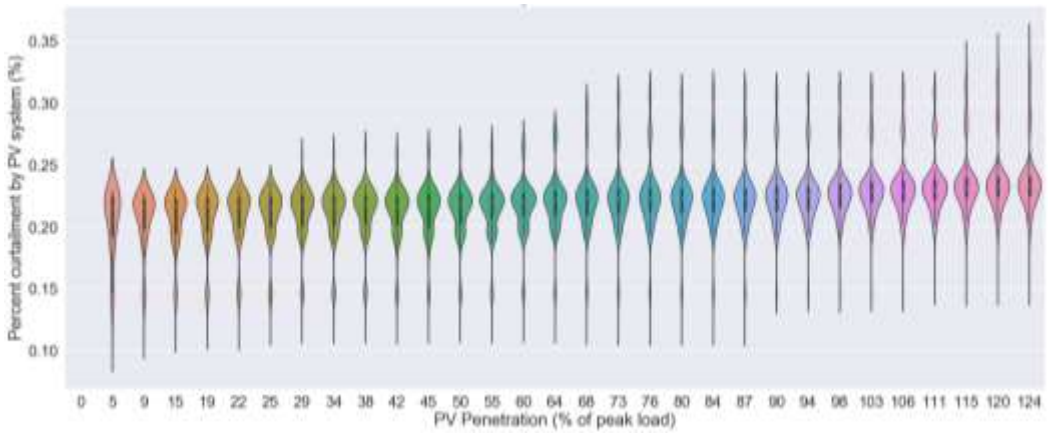


Figure 7. A violin plot showing the distribution of curtailment among all DPV systems on the feeder when autonomous volt-var controls are used as a function of penetration level for Feeder A with PV installations located far from the substation. The white dot in the center of each "violin" represents the median curtailment; the thick black bar the interquartile range (IQR); the thinner black lines show -1.5 IQR and + 1.5 IQR; and the full shape of the violin shows the full distribution of curtailment experienced by individual DPV systems. The colors in this case only represent penetration level.

[2] https://towardsdatascience.com/violin-plots-explained-fb1d115e023d

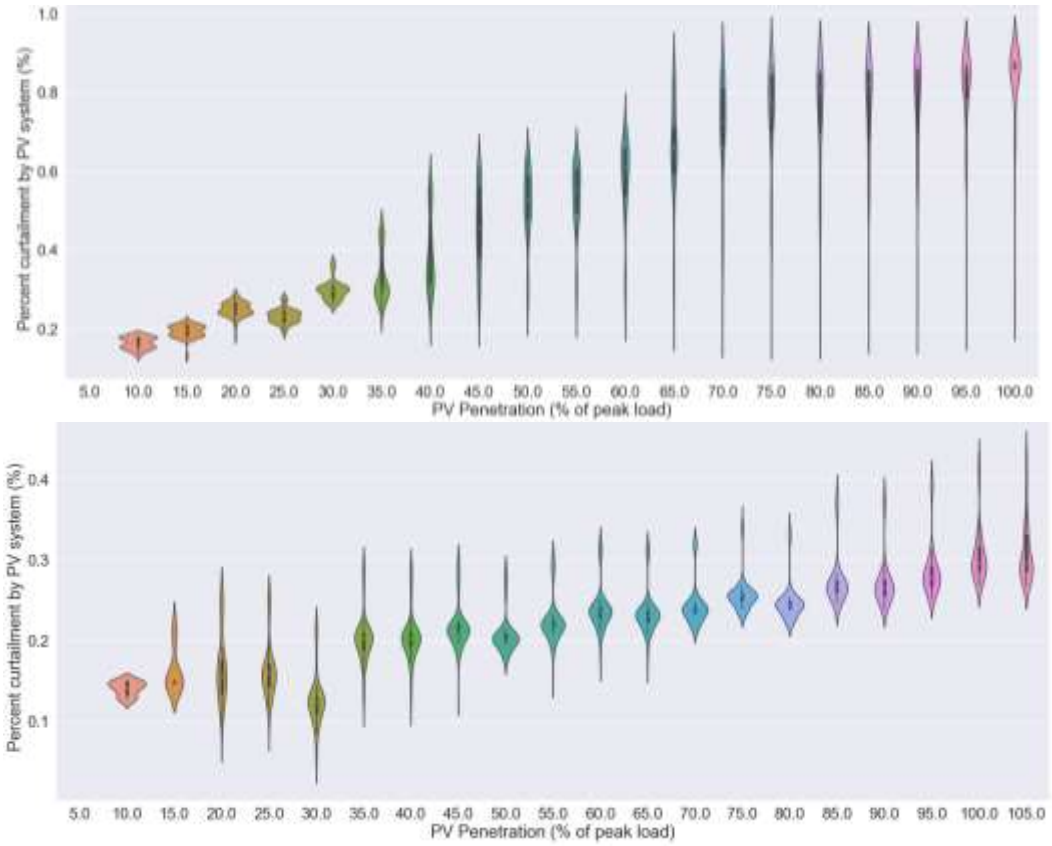


Figure 8. Violin plots showing the distribution of curtailment amon all DPV systems on the feeder when autonomous volt-var controls are used as a function of penetration level for Feeder B when PV systems are located far from the substation (top) and close to the substation (bottom). The white dot in the center of each "violin" represents the median curtailment; the thick black bar the interquartile range (IQR); the thinner black lines show -1.5 IQR and + 1.5 IQR; and the full shape of the violin shows the full distribution of curtailment experienced by individual DPV systems. The colors in this case only represent penetration level.

Figure 9 shows how the total curtailment (total DPV output with the volt-var controller divided by the total output of all DPV units on the feeder with PF = 1) changes with penetration and placement. Total curtailment levels are less than or equal to 0.24% for all cases for Feeder A.

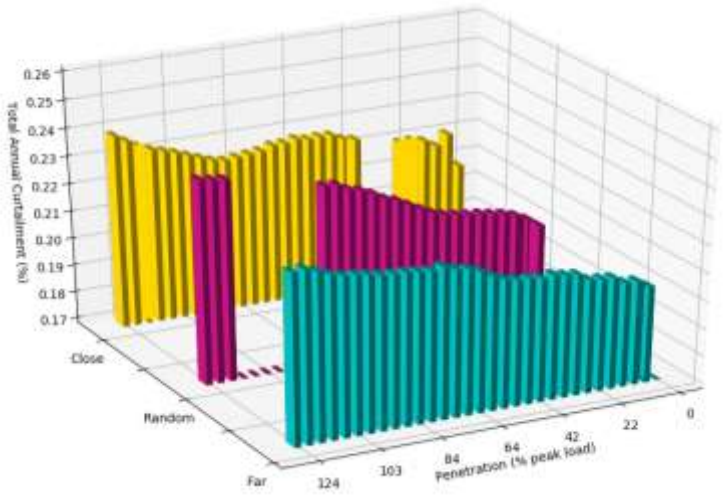


Figure 9. Total annual curtailment with volt-var for Feeder A for each of the placements as a function of penetration level. Missing bars are simulations that did not converge. Total curtailment levels are less than or equal to 0.24% for all cases. The colors and second x-axis indicate curtailment in the limiting placement for each PV location scenario: yellow = close to the substation, pink = randomly distributed, far = far from the substation.

The impact of DPV deployment on energy losses in each feeder with volt-var and DERMS controls is shown in Figure 10. This figure shows the change in losses compared to the case with no DPV generation. The losses decrease initially compared to the no-PV case but can rise above the original losses at high penetration levels. For some placements, the DERMS has a larger impact on energy losses for both feeders than the volt-var controls.

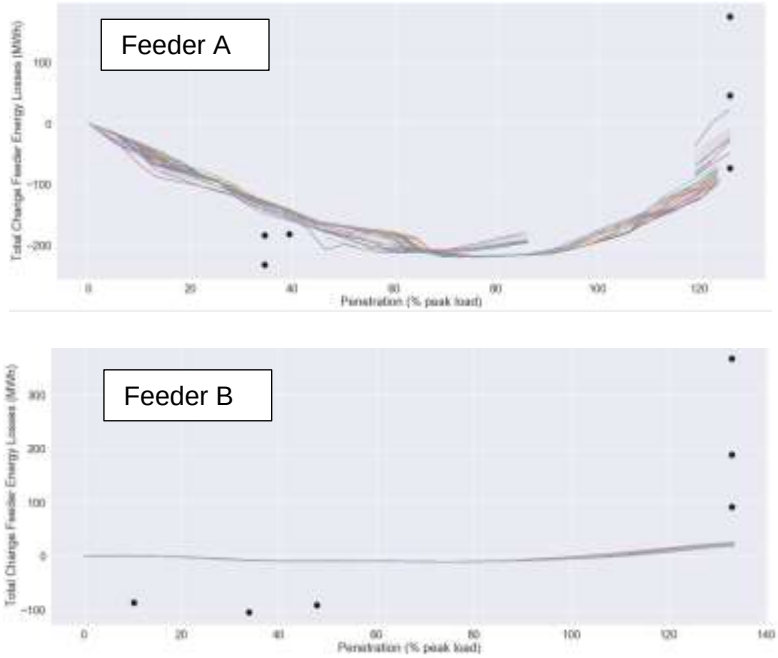


Figure 10. Change in total energy losses in the distribution system (lines, transformers, etc.) with volt-var controls (solid lines) or DERMS (black dots) compared to the case with no DPV generation for each feeder. The different-colored solid lines represent the 30 different PV locations (3 placements, 10 deployments per placement).

Finally, we examine how the number of grid capacitor switch operations is impacted by increasing DPV penetrations with volt-var controls or DERMS controls. While device lifetime generally depends on many factors other than the number of operations, as discussed above, large increases in device switching could lead to early wear-out and associated replacement costs, and how device operations are influenced by DPV penetration has been a question of interest to utilities and regulators (Palmintier, Broderick, et al. 2016). The two feeders studied here have voltage regulation provided by switched capacitor banks, rather than voltage regulators or LTCs. Thus, we are analyzing the changes in number of times that the capacitor bank switches themselves operate. The capacitors are all voltage-controlled, so they will switch more often in response to more frequent voltage changes; the number of switching operations for capacitor banks with time- or temperature-based controls will not change as DPV systems are deployed.

The results are shown in Figure 11. The number of capacitor switch operations are shown for the close-to-the-substation placement for both feeders, but the results are similar in terms of the number of operations and how the operations change with penetration level for the other placements. For Feeder A, the number of operations increases significantly compared to the no-PV case with volt-var controls for three out of five of the capacitor switches. With the DERMS control, the number of switching operations increases at high penetrations for all the capacitors, but the increase is smaller for some of the capacitors than for the volt-var case. For Feeder B, the use of volt-var actually reduces or does not impact the number of capacitor operations compared to the no-PV case for both capacitors, but the DERMS increases the number of operations significantly for the high-penetration case. As mentioned above, this may not fully capture all capacitor switch operations if the device delays are set to be less than one minute (the resolution of this study).

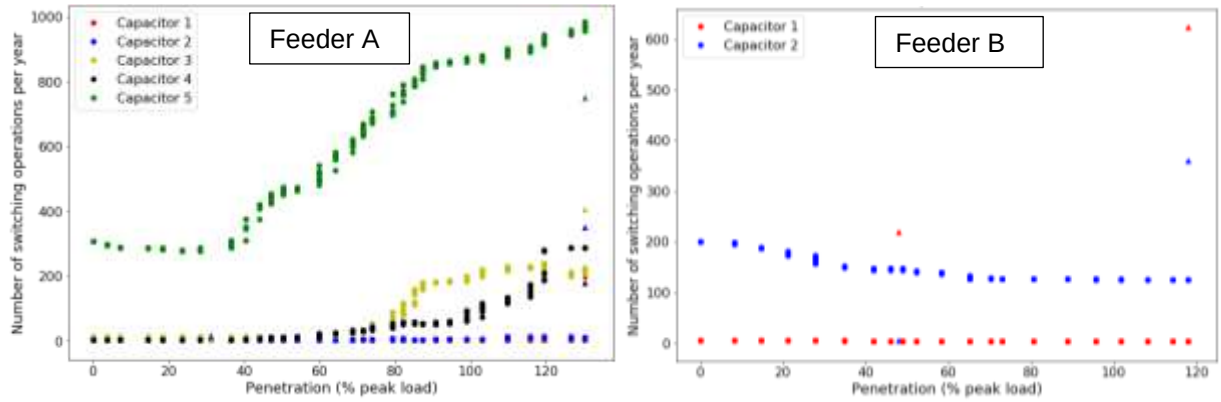


Figure 11. Number of capacitor switch operations per year for the close to the substation placement. The triangles indicate the number of operations with DERMS for each capacitor, while the dots show operations with volt-var controls. The number of annual capacitor switching operations was higher for the DERMS case at high penetration levels for most of the capacitors, but the total number of annual operations in all cases was still low.

For capacitor switches with a 30,000-operation lifetime of 30 years, one would expect approximately 1,000 operations/year to be tolerable. Even for the highest penetration on Feeder A, the maximum number of capacitor switch operations for the most affected capacitor is only around 1,000/year, with the number of operations for all other switches being below 400/year

and 200/year with volt-var for Feeder A and B, respectively, and below 400/year and 650/year with DERMS for Feeder A and B, respectively. The number of operations could potentially also be reduced by adjusting the switching delay. Thus, for these feeders and capacitors, the deployment of DPV systems and use of advanced inverters may not measurably influence device wear-out or replacement costs.

Other Curtailment Risk Considerations

In the DERMS scenario, which involves coordination of DPV output via a communications channel, the communications to the inverter may be lost. Different approaches are taken to deal with this loss of communication in practice, including defaulting to autonomous/local controls or curtailing the DPV output entirely until communications are restored. We do not explicitly model this communications downtime. PG&E's EPIC 2.03 study found that communications uptime was greater than 95% for 85% of the days in the year, with many days having a higher percentage of uptime (PG&E 2018). For a particular interconnection scenario, communications uptime can be applied in post-processing to better understand curtailment risk.

There is also a risk that when DERMS is being used to ensure the DPV output does not cause violations on the distribution system, the DPV output will be reduced to zero while maintenance is being performed on the DERMS or if the DERMS goes down for any reason. Due to the limited number of DERMS deployments so far and the evolving nature and maturation of the technology, there is little data available on what this downtime could be today. However, downtime for maintenance or updates of the control system itself has not been mentioned as a major challenge in reports on early advanced inverter, DERMS, and DER aggregation pilots (Cook et al. 2018; Ardani, O'Shaughnessy, and Schwabe 2018; PG&E 2019).

As mentioned above, this analysis focuses on the mitigation of steady-state voltage and thermal violations on the distribution system. Any curtailment associated with protection or protection coordination issues or intended to avoid reverse power at the substation is not reflected in our study (i.e., protection is not considered part of our controls).
Finally, if there are only some DPV systems on a feeder participating in the DERMS scheme, this can affect the amount of curtailment for each generator. This is the subject of future research but is not considered in this analysis.

Cost of Curtailment and Losses

The cost of curtailment itself for the volt-var case is expected to be modest. For example, the weighted average price of wholesale electricity from the SP-15 price hub applicable in Southern California where these feeders are located was 50.62 ¢/MWh in 2018 (EIA 2019). The maximum amount of total curtailment for Feeder A at 131% of peak load was calculated to be 44,712 kWh. Thus, if operating under wholesale market prices, this cost could be as low as $2,263/year. Using NREL's System Advisor Model (SAM), we find that amount of curtailment would have a very small impact on overall monthly bill savings for residential or small commercial installation, with less than $1 reduction in monthly electricity bill savings under current policy in Southern California with 0.45% annual curtailment.

However, for the high-penetration scenario on Feeder B with the DERMS control where curtailment is higher (15% to 44%), the impacts for customers increase, with a reduction in

monthly bill savings ranging from $12 to more than $40/month. At 32% curtailment (random DERMS, high-penetration scenario), lost revenue due to total curtailment on the feeder if selling power at the SP-15 wholesale price could exceed $70,000. The exact impact varies system-to-system—as well as regionally—and will evolve over time as PV system costs, electricity prices, and policies change. Additionally, small residential and commercial PV systems are not currently selling power into wholesale markets. The simplified examples above serve only to give a sense of the magnitude of this potential cost for comparison with the cost of upgrades and other distribution system impacts.

For energy losses on the feeder, using the SP-15 average wholesale price of 50.62 ¢/MWh and the data from Figure 10, the cost associated with increased line losses is below $180/year for both feeders; the maximum savings due to reduced line losses would be $125/year. Thus, DPV deployment and the choice of an advanced inverter control scheme (either DERMS or autonomous volt-var) is expected to have a minimal impact on the overall cost of losses for these two feeders.

REVERSE POWER FLOW

Reverse power flow at the feeder head occurs at between 20% and 30% of peak load for both feeders, depending on the location of the DPV systems, with instances of reverse power flow rising as penetration levels increase further. Reverse power flow can trigger the need to enable co-generation or bi-directional modes on equipment such as regulators, substation transformers, or reclosers that were originally set up to operate on the assumption of one-way power flow. These upgrades can be expensive, with reported costs ranging from $60,000 to $145,000 (“Distribution Grid Integration Unit Cost Database | Solar Research | NREL” 2018). Reverse power flow can also trigger protection coordination or other issues at the substation; in some cases, utilities limit the amount of DPV generation installed or the output (i.e., via curtailment) to avoid any reverse power flow at the substation. While this is an important consideration, the impacts and costs of reverse power flow on protection and an understanding of what upgrades are required to manage reverse power flows are areas of active research and are beyond the scope of our study.

VI. CONCLUSION AND FUTURE WORK

In this study, we explore the potential distribution grid costs associated with integrating residential and small commercial DPV systems onto the grid using traditional upgrades, autonomous volt-var controls, and DERMS controls by modeling two real feeders in Southern California with different characteristics. We compare the up-front costs, potential curtailment of DPV output, line losses, number of capacitor switch operations, and effectiveness at expanding the hosting capacity for each of these integration solutions. The intent is to enhance understanding of the overall cost drivers at different penetration levels and help establish a framework for considering the total set of potential costs to all parties (utilities, DPV owners, and ratepayers) as a function of penetration level.

We find that a DERMS can allow for high DPV penetration while mitigating voltage and thermal overloads but results in high levels of DPV curtailment at penetration levels above 100% of peak load. DERMS today also still have a somewhat high up-front cost. Autonomous volt-var controls, on the other hand, have no up-front cost and result in less DPV curtailment. However,

our analysis finds they are not able to mitigate all voltage and thermal problems on the circuit and would require reinforcement with traditional upgrades to reach higher penetration levels. The exact amount of hosting capacity expansion achieved by volt-var varies significantly depending on where the PV system is located on the feeder.

Similarly, we find that curtailment with volt-var and, in particular, the DERMS controls varies significantly depending on the location (placement) of the DPV systems. The DERMS controls result in higher variability in the level of curtailment among the DPV generators than in the volt-var scenario. While traditional upgrades do not involve any curtailment risk for the DPV generators, they can have a high up-front cost. There is a penetration level up to which the curtailment cost/risk for the DPV units is less than the cost of these upgrades and a point beyond which curtailment becomes uneconomical (for example, for many of the DERMS scenarios at > 100% peak load for Feeder B) and additional grid reinforcement may be more economical. However, the exact point at which this occurs will depend on how costs are allocated among DPV owners, the utility, and ratepayers; the business models of the DPV system owners and electricity pricing; and the characteristics and resources of DPV system owners.

While the number of modeled capacitor switch operations increase for some capacitor banks as penetration levels increase for each feeder, the absolute number of capacitor switching operations is likely not large enough to cause early device wear-out. Additional study is needed to understand how the number of voltage regulator or LTC operations could be impacted for feeders using those devices for voltage regulation. We find that the distribution system losses may increase or decrease with DPV deployment depending on the feeder and penetration level, but the overall costs or savings of changes in losses for these two feeders is likely to be small.

Ultimately, which solution is most appropriate for expanding the hosting capacity will depend on the specific circumstance, including current penetration levels, how much the PV hosting capacity needs to be expanded (e.g. expected or desired PV system deployment on a given feeder), any capital and/or regulatory constraints for a specific utility, and the degree to which a DERMS system could utilize existing communications infrastructure. This paper provides initial assessment of trade-offs between different solutions at different penetration levels in order to inform decisions about which may be most appropriate in a given scenario. Future work is needed to assess a larger number of feeders and understand the extendibility of these results to a broader set of geographic locations. Additionally, sensitivity analyses are required to understand how the effectiveness of DERMS and volt-var controls vary as the density of monitoring points, number of devices controlled, zones of control or aggregation, curtail and trip thresholds, and other factors are considered. Scalability of the DERMS approach, including potential costs associated with additional communications infrastructure, new cybersecurity requirements, etc., is also warranted as the technology matures and larger numbers of devices may be controlled. Additional controls approaches, including deterministic controls as opposed to optimization algorithms, are also of interest for future study. More data is also required to better understand the potential O&M costs associated with both traditional upgrade and emerging DERMS integration solutions. Finally, future work could explore refined DERMS optimization approaches that might enable maintaining effective increases in hosting capacity with lower levels of DPV curtailment.

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APPENDIX

Description of the DERMS algorithm used in the study:

Here we describe in more detail the optimal energy dispatch control algorithm to solve the optimal active power and reactive power outputs from PV inverters to maximize PV generation while not violating voltage and thermal operational limits.

A. Objective Function

The objective of each DERMS is to minimize active power curtailment and reactive power output from all controlled PV inverters, and it is formulated as:

$$\min_{p_j^t, q_j^t} f^k(\mathbf{x_k^t}) = \sum_{j=1}^{NPV_k} c_P^k \cdot \left(p_{j,k}^{t,max} - p_{j,k}^t\right)^2 + c_Q^k \cdot \left(q_{j,k}^t\right)^2 \quad (1)$$

where, $\mathbf{x_k^t} = \left\{p_{j,k}^t, q_{j,k}^t,\ j = 1, \ldots, NPV_k\right\}$, and $p_{j,k}^t$ and $q_{j,k}^t$ are actual active power output and reactive power output from the jth PV inverter at time t. NPV_k is the total number of distributed PV inverters controlled by DERMS-k. $p_{j,k}^{t,max}$ is the maximum active power output that can be generated from the jth PV inverter at time t. c_P^k and c_Q^k are constant coefficients, and typically $c_P^k \gg c_Q^k$.

The feasible solution set for p_j^t and q_j^t is defined as

$$\chi_k^t = \begin{cases} 0 \le p_{j,k}^t \le p_{j,k}^{t,max} \\ \left(p_{j,k}^t\right)^2 + \left(q_{j,k}^t\right)^2 \le S_{j,k}^2 \\ \left(q_{j,k}^t\right)^2 \le 0.44^2 \cdot S_{j,k}^2 \end{cases} \quad (2)$$

where, S_j is the inverter size for the jth PV inverter.

B. Constraints

Each DERMS tries to maintain the voltages and currents at its measurement nodes to meet operation requirements as defined in (3) and (4). Currents should not exceed their thermal limits. Voltages should follow ANSI service voltage limit[3], i.e. $\pm$ 5% deviation from the nominal value.

$$\underline{v} \le |v_i^t| \le \overline{v}, \ \forall i \ \in N_{measure}^k \quad (3)$$

$$|I_i^t| \le \overline{I}_i, \ \forall i \ \in N_{measure}^k \quad (4)$$

where, v_i^t and I_i^t are respectively the voltage at the ith measurement node at time t, $\overline{I_k}$ is the upper current limit for the kth branch. $\underline{v}$ and $\overline{v}$ are the lower and upper voltage limits for the targeted operational range (e.g. 0.95 p.u. and 1.05 p.u.). $|\cdot|$ is the magnitude of a complex value. $N_{measure}^k$ is the total number of measurement nodes in DERMS-k.

C. Algorithm

Finally, the optimization problem solved in DERMS-k can be formulated as the following convex optimization problem.

$$\begin{aligned} \min_{p_j^t, q_j^t \in \chi^t} \ & f^k(\mathbf{x_k^t}) \\ s.t. \quad & g_i^k(\mathbf{x_k^t}) \le 0, \\ & h_i^k(\mathbf{x_k^t}) \le 0, \\ & l_i^k(\mathbf{x_k^t}) \le 0, \qquad and \ i \in \mathcal{N}_k \end{aligned} \quad (5)$$

The primal dual gradient algorithm is used to solve the problem. And, power set points $\mathbf{x_k^t}$ are updated every time step, as

$$\mathbf{x_k^t} = \text{Proj}\left\{\mathbf{x_k^{t-1}} - \alpha_x \cdot \nabla_x \mathcal{L}(\mathbf{x_k^{t-1}}, \overline{\boldsymbol{\mu}}_\mathbf{k}^{\mathbf{t-1}}, \underline{\boldsymbol{\mu}}_\mathbf{k}^{\mathbf{t-1}}, \boldsymbol{\gamma}_\mathbf{k}^{\mathbf{t-1}})\right\} \quad (6)$$

$$where, \begin{cases} \overline{\boldsymbol{\mu}}_\mathbf{k}^{\mathbf{t}} = \text{Proj}\left\{\overline{\boldsymbol{\mu}}_\mathbf{k}^{\mathbf{t-1}} + \alpha_{\overline{\mu}} \cdot \nabla_{\overline{\mu}} \mathcal{L}(\mathbf{v_k^{t-1}} | \mathbf{x_k^{t-1}})\right\} \\ \underline{\boldsymbol{\mu}}_\mathbf{k}^{\mathbf{t}} = \text{Proj}\left\{\underline{\boldsymbol{\mu}}_\mathbf{k}^{\mathbf{t-1}} + \alpha_{\underline{\mu}} \cdot \nabla_{\underline{\mu}} \mathcal{L}(\mathbf{v_k^{t-1}} | \mathbf{x_k^{t-1}})\right\} \\ \boldsymbol{\gamma}_\mathbf{k}^{\mathbf{t}} = \text{Proj}\left\{\boldsymbol{\gamma}_\mathbf{k}^{\mathbf{t-1}} + \alpha_\gamma \cdot \nabla_\gamma \mathcal{L}(\mathbf{v_k^{t-1}} | \mathbf{x_k^{t-1}})\right\} \end{cases} \quad (7)$$

D. Implementation

The implementation architecture to realize the proposed distributed DERMS control is shown in Figure 3 of the main body. There are three components in the DERMS: 1) Aggregation—

[3] ANSI C84.1-2016, "American National Standard for Electric Power Systems and Equipment – Voltage Ratings (60 Hz)."

registers all PV systems under control and acquires the information of PV locations and sizes, 2) Monitoring—measures real-time voltages, currents, and powers at all sensor locations, and measures real-time PV inverter outputs, and 3) Optimal Control—solves the optimal PV inverter power set points. There are two options to implement DERMS control: 1) Optimal control solves the power set points for PV inverters every time step; 2) Optimal control component is only implemented when there is violation observed at the sensing locations. Otherwise, PV systems receive no external control signals, and PV inverters are generating the maximum PV power output. To date, have focused on studying the second option; later, we will compare the results for both options.